Moving to a SQL Server

Lecture 1: IMDb, PostgreSQL, and indexes

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Today

Step	Working result
Get IMDb	Two checked gzip archives
Start PostgreSQL	A successful client query
Import CSV	Tables with keys and indexes
Inspect a plan	Actual server work

File Database and Server Database

SQLite	PostgreSQL
Python opens a file	Python connects to a process
sqlite3 driver	Psycopg driver
Database file	Database name and connection

The server can run in the same Codespace.

IMDb Inputs

Archive	Useful fields
title.basics.tsv.gz	Identifier, title, year, runtime, genres
title.ratings.tsv.gz	Identifier, mean rating, vote count

The identifier connects the two files.

Exercise: Get the Files

```
cd chapter-15-moving-to-a-sql-server/code
bash download-imdb.sh
git check-ignore -v data/title.basics.tsv.gz
```

Result: two valid archives and a matching ignore rule.

Exercise: Start PostgreSQL

```
bash install-postgres.sh
bash start-postgres.sh
```

Setting	Exercise value
Address and port	127.0.0.1:5433
Database and role	imdb, student

Exercise: Verify the Connection

```
export DATABASE_URL=postgresql://student@127.0.0.1:5433/imdb
psql "$DATABASE_URL" -c \
  "SELECT current_database(), current_user;"
```

Expected: imdb and student.

Exercise: Prepare the Sample

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python prepare_csv.py --start-year 1975 --end-year 1999 --limit 0
```

Result: titles.csv and matching ratings.csv.

Two Tables, One Identifier

titles	ratings
title_id: primary key	title_id: primary and foreign key
primary_title, start_year	average_rating
runtime_minutes, genres	num_votes

Each rating belongs to one imported title.

Exercise: Import and Check

```
python import_imdb.py
python check_setup.py
psql "$DATABASE_URL" -f lecture-01-checks.sql
```

Result: printed row counts, index names, and a query plan.

Indexes for Actual Queries

Index	Query pattern
start_year, primary_title	One year, ordered titles
primary_title	Exact title equality
num_votes descending	Most-voted titles

Ordinary title indexes do not solve arbitrary substring search.

Read the Plan

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT title_id, primary_title
FROM imdb_course.titles
WHERE start_year = 1994
ORDER BY primary_title;
```

The chosen plan depends on the data and query.

Ready for the Dashboard

The server is reachable.

The tables have verified row counts.

The data layer can now return query results.

Keep the server port local; next meeting adds the web interface.